Assignment No 10

Aim : Design a model to detect the Fake News from Multimodal data.

Theory:

Fake News Detection is an important application of Artificial Intelligence and Machine Learning that aims to identify whether a news article or social media post contains **true or misleading information**. With the rapid growth of digital media, large amounts of information are shared through websites, social media, images, and videos. Therefore, automated fake news detection systems can help identify potentially unreliable content.

Multimodal data refers to data that contains information from **multiple modalities or sources**, such as **text, images, audio, and video**. A multimodal fake news detection model combines information from different types of data instead of relying only on the written text.

For example, a social media post may contain:

- News headline or article text
- An image
- Caption
- Source information
- Video or audio

A multimodal model analyzes these different inputs together to determine whether the content is likely to be **Real News or Fake News**.

Need for Multimodal Fake News Detection

Traditional fake news detection systems often analyze only the **textual content** of an article. However, fake news can also use manipulated images, misleading captions, or unrelated photographs.

For example, an article may contain believable text but use an **old or unrelated image** to create a misleading impression. A multimodal model can analyze both the text and image to identify such inconsistencies.

Components of the Model

A multimodal fake news detection system can contain the following components:

1. Text Processing

The textual content such as headlines, articles, and captions is processed using Natural Language Processing (NLP).

Common preprocessing steps include:

- Removing unnecessary characters.
- Converting text to lowercase.
- Removing stop words.
- Tokenization.
- Removing unnecessary punctuation.
- Converting text into numerical features.

Text can be represented using techniques such as **TF-IDF, Word Embeddings, Word2Vec, BERT, or other language models**.

2. Image Processing

Images associated with news content can be analyzed using **Computer Vision** techniques.

Features can be extracted using models such as:

- CNN
- ResNet
- VGG
- EfficientNet

These features help the model understand the visual information present in the image.

3. Feature Fusion

The features obtained from text and images are combined into a single representation. This process is called **multimodal feature fusion**.

For example:

Text Features + Image Features → Combined Features

The combined information is then provided to a classification model.

4. Classification

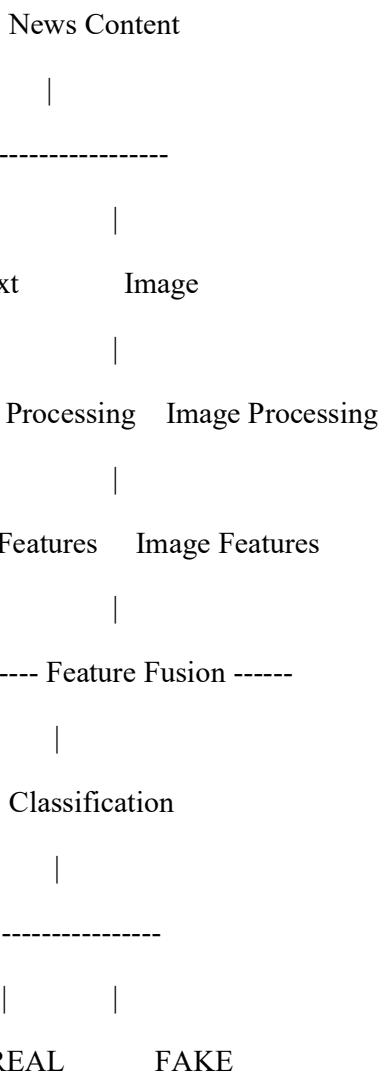
The final classifier predicts whether the news is:

- **Fake**
- **Real**

Classification can be performed using algorithms such as Logistic Regression, Random Forest, SVM, Neural Networks, or Deep Learning models.

Basic Architecture

A simple multimodal fake news detection system can be represented as:



Steps Involved

1. **Collect multimodal dataset** containing news text and associated images.
2. **Clean and preprocess the text data.**
3. **Extract features from the text.**
4. **Preprocess the images.**
5. **Extract visual features from images.**

6. **Combine text and image features.**
7. **Split the dataset into training and testing sets.**
8. **Train the classification model.**
9. **Evaluate the model using appropriate performance metrics.**
10. **Predict whether new multimodal news content is real or fake.**

Machine Learning and Deep Learning Techniques

Different algorithms can be used for multimodal fake news detection:

- **TF-IDF + Logistic Regression**
- **TF-IDF + SVM**
- **Random Forest**
- **CNN for image features**
- **BERT for text features**
- **CNN + LSTM**
- **BERT + CNN**
- **Transformer-based multimodal models**

For a **beginner-level practical**, a simple implementation can use **TF-IDF for text features** and image features obtained using a pretrained CNN, followed by a classification algorithm.

Evaluation Metrics

The performance of a fake news detection model can be evaluated using:

Accuracy:
Measures the percentage of correctly classified news items.

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

Precision:
Measures how many items predicted as fake are actually fake.

Recall:
Measures how many actual fake news items were correctly detected.

F1-Score:
It combines precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

A **confusion matrix** can also be used to visualize correct and incorrect predictions.

Advantages

- Uses information from multiple data sources.
- More comprehensive than text-only fake news detection.
- Can identify inconsistencies between text and images.
- Useful for social media monitoring.
- Can process large amounts of digital content automatically.

Limitations

- Requires multimodal datasets containing different types of data.
- Feature extraction can be computationally expensive.
- High-quality labeled datasets can be difficult to obtain.
- Manipulated or highly sophisticated content may still be difficult to detect.
- Combining different modalities increases model complexity.

Applications

Multimodal fake news detection can be used for:

- Social media monitoring.
- News verification.
- Online content moderation.
- Misinformation detection.
- Fact-checking assistance.
- Digital journalism.
- Detection of misleading advertisements.
- Monitoring viral misinformation.

Conclusion

A **multimodal fake news detection model** combines information from multiple sources such as **text and images** to determine whether news content is genuine or fake. By using **NLP, computer vision, feature extraction, and machine learning/deep learning**, the system can provide a more comprehensive approach to misinformation detection than a text-only model.

Code:

```
import numpy as np

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.preprocessing import StandardScaler

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, classification_report

texts = [

    "Breaking news: new policy announced",

    "Fake news: celebrity scandal",

    "Economy is growing steadily",

    "Fake: aliens discovered in desert",

    "Sports event draws large crowd",

    "Fake report about virus outbreak"

]

metadata = [

    [0.9, 0.2],

    [0.1, 0.8],

    [0.8, 0.4],

    [0.2, 0.9],

    [0.85, 0.5],
```

```
    [0.3, 0.7]]

labels = [1, 0, 1, 0, 1, 0]

vectorizer = TfidfVectorizer(max_features=50)text_features
=vectorizer.fit_transform(texts).toarray()

scaler = StandardScaler()

meta_features = scaler.fit_transform(metadata)

X = np.hstack((text_features, meta_features))

y = np.array(labels)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

model = LogisticRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

print("Classification Report:\n", classification_report(y_test, y_pred))

new_text = ["Fake news: politician involved in scandal"]

new_meta= [[0.2, 0.7]]

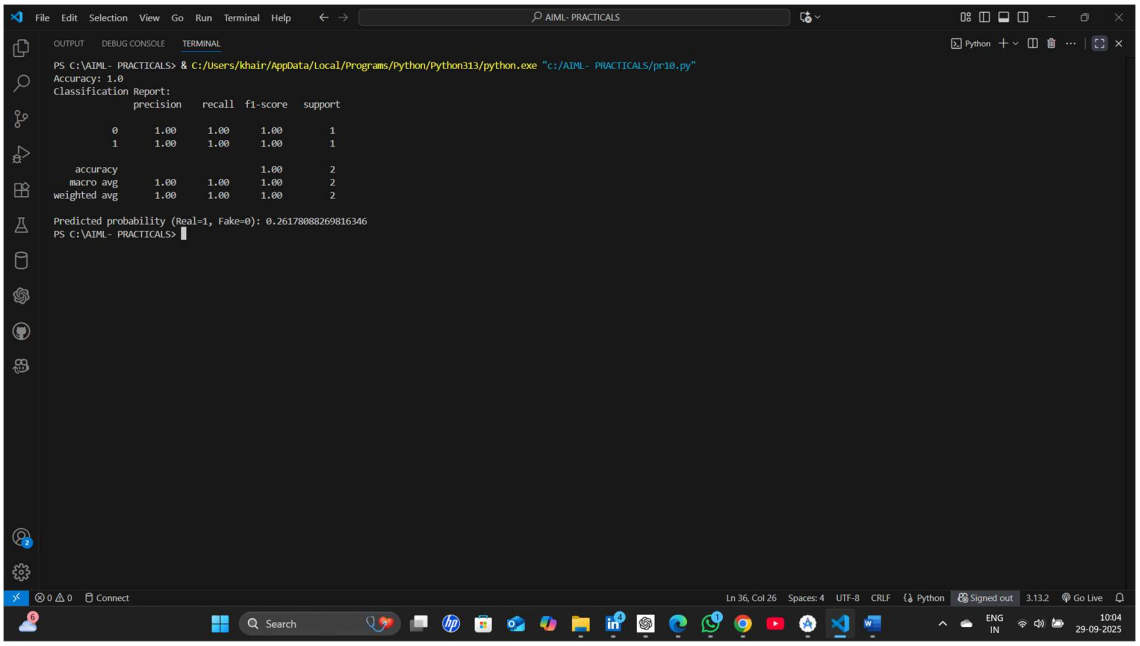
new_text_features = vectorizer.transform(new_text).toarray()

new_meta_features = scaler.transform(new_meta)

new_X = np.hstack((new_text_features, new_meta_features))

pred = model.predict_proba(new_X)[:,1]

print("Predicted probability (Real=1, Fake=0):", pred[0])
```


File Edit Selection View Go Run Terminal Help
AIML- PRACTICALS
OUTPUT DEBUG CONSOLE TERMINAL
Python
PS C:\AIML- PRACTICALS> & C:/Users/khair/AppData/Local/Programs/Python/Python313/python.exe "c:/AIML- PRACTICALS/pr10.py"
Accuracy: 1.0
Classification Report:
precision recall f1-score support
0 1.00 1.00 1.00 1
1 1.00 1.00 1.00 1
accuracy 1.00 2
macro avg 1.00 1.00 1.00 2
weighted avg 1.00 1.00 1.00 2
Predicted probability (Real=1, Fake=0): 0.2617808826981634
6
PS C:\AIML- PRACTICALS>
Connect
Ln 36, Col 26 Spaces: 4 UTF-8 CRLF Python Signed out 3.13.2 Go Live
Search
ENG
IN
10:04
29-09-2025